

Operation System (OS) is a system software, that operates the system.

- Interface between user and hardware
- Memory Management.
- CPU scheduling

Examples: MacOS, Linux, Windows, Android, IOS.

Q. What is an Operating System?

Lecture-2

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⇒ An Operating System (OS) is a ~~pro~~ program that acts as an intermediary between a computer user and the computer hardware.

⇒ OS is software that controls the operation of computer and directs the processing of programs.

Goal: i) Execute user program

ii) Make the computer system convenient and user-friendly to use.

iii) Use computer hardware in an efficient manner.

⇒ Computer System Structure: There are 4 main components:

① Hardware: Provides basic computing resources. @.
Example: CPU, Memory, I/O devices.

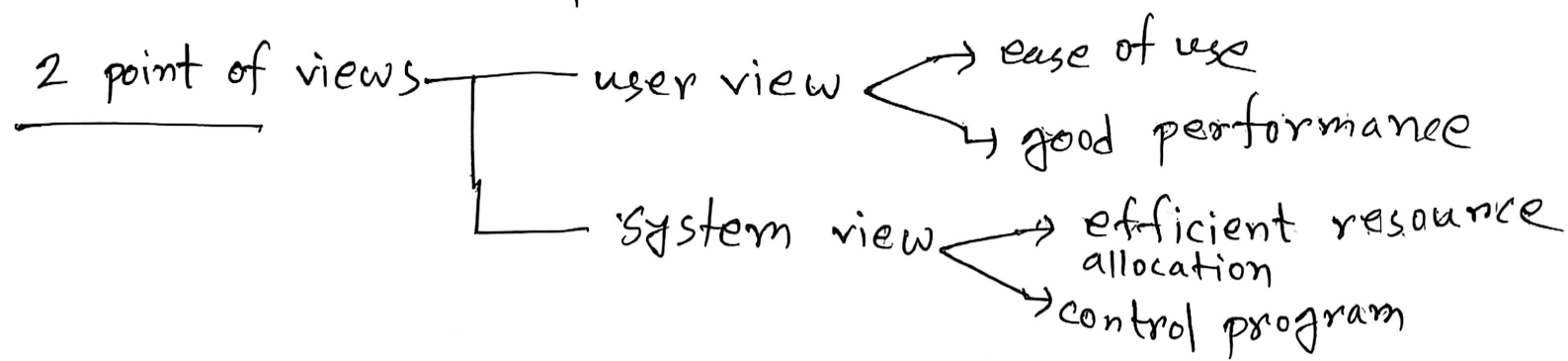
② Operating System: Controls and coordinates use of hardware among various applications and users.
Example: macOS, Windows, Linux

③ Application Program: Define how system resources are used to solve users computing problems.
Example: Compilers, web browsers, video games etc.

④ Users: Entities interacting with the system.
Example: People, machines, other components.

System Perspective

What an OS does depends on the user's point of view:



⇒ Single-User System: Prioritize convenience, ease of use and good performance; resource utilization is not primary concern.

2. Shared Computers: Focus on maximizing resource sharing so all concurrent users get fair access.

3. Workstations: Balance individual usability with sharing centralized resources from network servers.

4. Handheld/Mobile Devices: Focus on battery conservation and user responsiveness under limited hardware resources.

5. Embedded System: Found inside cars, appliances or machinery. Focused entirely on background operations.

Formal Definitions of OS Functions

Resource Allocator: Manages system resources and resolves competing demands fairly and efficiently.

Control Program: Controls execution of programs to prevent errors and improper use of the computer

Kernel: The primary Software components that remains loaded and actively running in memory at all times

Middleware: A set of software frameworks that provide additional services to application developers.

Bootstrap / Bootloader

Startup Role: A small initialization program loaded automatically when a computer powers up or reboots.

Storage Location: Stored permanently in non-volatile memory chips (ROM or EPROM), commonly known as firmware.

Task Executed: 1. Initializes hardware components across the board

2. Locates and loads the OS kernel into main memory to launch the OS.

Steps: 1. $\Rightarrow$ start up operation

$\Rightarrow$ CPU on $\rightarrow$ ROM এর সর্ব প্রথম Instruction থাকে।

$\Rightarrow$ ROM এর মধ্যে software হিসাবে থাকে firmware

$\Rightarrow$ Firmware এর মধ্যে থাকে BIOS (Basic Input Output System)

$\Rightarrow$ BIOS এর code থাকে এবং initialize রত্ন active করে।

$\Rightarrow$ OS secondary memory এ থাকে এবং H.D. থেকে OS এর copy RAM-এ আনে।

Computer System Organization

Hardware Interconnection: One or more CPUs together with various device controllers connect a single common bus providing access to shared memory.

Concurrent Execution: Hardware ^{devices} and processors operate at the same time, competing for available memory access cycle.

Types of Operating System

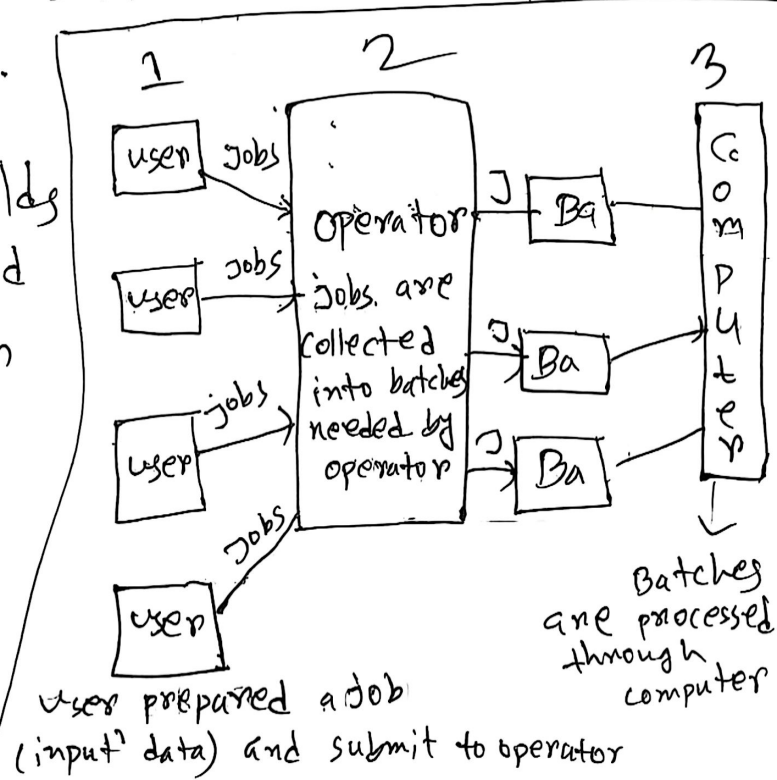
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26.07.2026

Batch Operating System: A system designed for non-interactive tasks where jobs are executed automatically without direct user intervention.

Batching Concept: To speedup overall execution, individual user jobs requiring similar software or setup are grouped together into batches.

How it works: The system holds queued jobs in memory and processes them sequentially in First-Come, First-Served (FCFS) order.

When a job is completed,



its allocated memory is freed and its output is redirected to an output spool for later printing.

Examples

(i) Telecom Billing: Batch processing monthly telephone call records to generate customer bills.

(ii) Bank Processing: Running overnight processing for international transfers or clearing accounts.

(iii) Corporate Payroll Systems: Calculating monthly salaries and printing employee pay slips.

Advantages

(i) Faster Group Execution: Same jobs in the batch are higher executed speed.

(ii) No user Overhead: Automated transitions mean the next job running instantly once the active job finishes.

(iii) Better CPU Utilization: Minimizes CPU idle time.

(iv) Task Partitioning: Large batch processes are divided into smaller sub-processes for better handling.

Disadvantages

(i) Difficult to debug

(ii) Infinite loop risk

(iii) Batch systems are costly.

Time-Sharing Operating System

Def:

A system model that allows multiple users at different terminals to access a single central computer at the same time.

Time Quantization: The CPU ~~processing~~ assigns an equal portion of time (called a time ~~quantum~~ slices) to each task so every process run smoothly.

$$U_1 (10 \text{ min}) / 7 \text{ min} / 4 \text{ min} / 1 \text{ min} / x$$

$$U_2 (15 \text{ min}) / 12 \text{ min} / 9 \text{ min} / 6 \text{ min} / 3 \text{ min} / x$$

$$U_3 (3 \text{ min}) / 0 \text{ min} / x$$

$$TQ = 3 \text{ min}$$

Characteristics: (i) Multi-user system

(ii) fixed time (iii) Fair CPU sharing

Multitasking: Each user receives a portion of CPU time making these systems effectively known as multi-tasking system.

Time-sharing is considered a logical step forward from basic ~~mult~~ multiprogramming.

Examples: Multics, Unix etc.

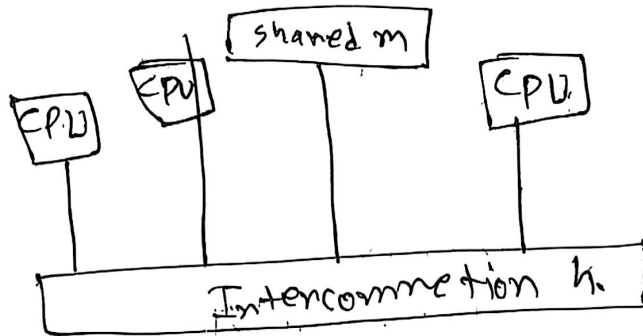
Distributed Operating System

Definition: A system whose individual components are spread across ~~different~~ networked computers that communicate by passing message. Example: Locus, AEGIS, etc

⇒ Interconnected autonomous computers run together, where each individual system has its own CPU and memory unit. This is called loosely coupled systems.
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Coupled Systems

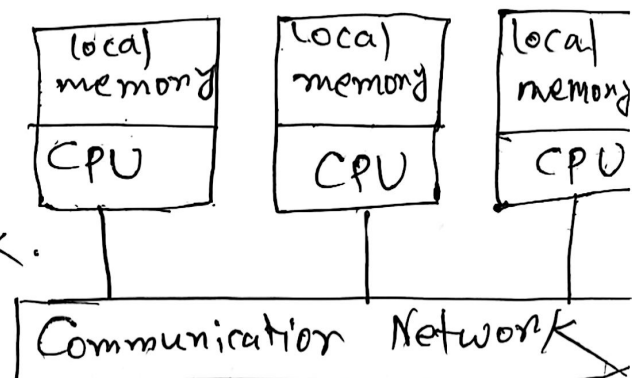
Tightly Coupled System: Processors share a single main memory and communicate directly over a high-speed bus. Examples: Parallel Processing, shared memory multiprocessors



Loosely Coupled Systems:

Independent computers have separate memory units and exchange messages over a network.

Example:



Network Operating Systems (NOS)

Definition: An operating system running on a network host that enables shared access to files, printers, security services and applications across a private network.

Examples: Windows Server 2003/2008, UNIX, Linux, macOS X, Novell NetWare, BSD.

Distributed OS vs Network OS

Feature	Distributed OS	Network OS
Definition	Multiple computers work together as one system	Multiple independent computers connected by a network
User view	Appears as a single Computer	Users know they are using different computers
System Image	Single System Image (SSI)	No single system image
Transparency	High; location of resources is hidden.	Low; users explicitly access remote resources
Operating System	Cooperating distributed OS across nodes	Each computer runs its own OS independently
Scalability	Excellent	Good, but manual administration needed
Administration	Appears Centralized	Each machine is also monitored separately
Examples	Amoeba, Plan 9, Sprite	Windows Server, Ubuntu Server, macOS X.

Mobile Operating System

Target Hardware: Designed specifically for handheld smartphones and tablets

Special features: Includes built-in support for hardware sensors like GPS and gyroscopes, allowing advanced apps such as augmented reality.

Connectivity: Connects using cellular data networks or IEEE 802.11 wireless (WiFi)

Market leaders: Apple iOS, Google Android.

Embedded Operating System

Definition: Specialized operating systems designed for computers built into larger machinery or consumer appliances.

Application: Vehicles, traffic signals, smart TVs, ATMs, flight controls, Point of Sale (POS), GPS receivers, digital cameras.

Real-Time Operating System (RTOS)

Definition: An operating system designed to guarantee that specific tasks complete within strict, predefined time limits.

Core Requirement: Operations must be produce correct logic within strict deadline constraints.

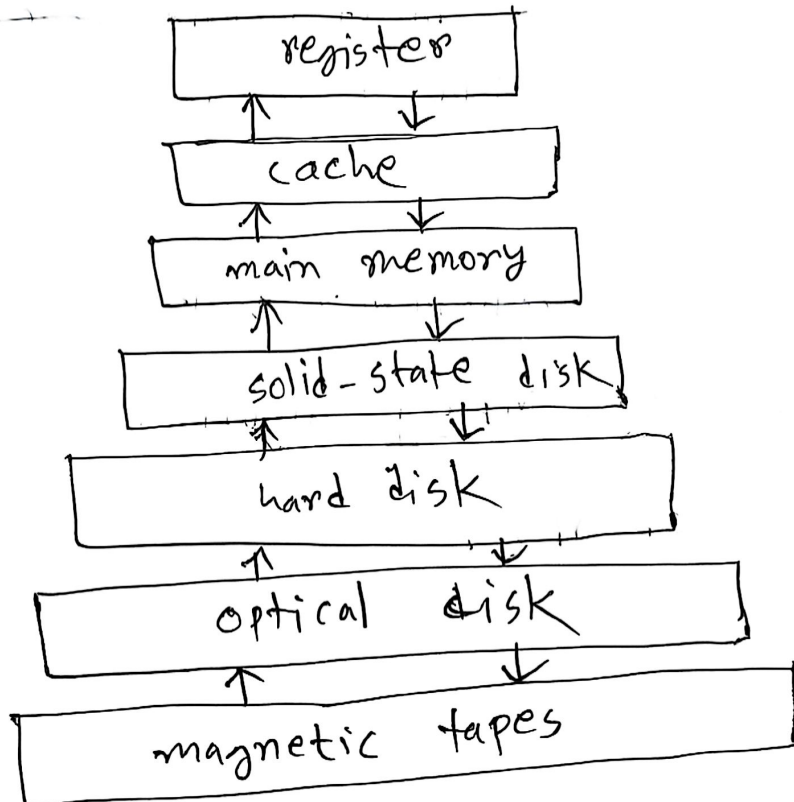
Applications: Industrial machinery, medical imaging, flight control, weapon control system, scientific experiments.

Storage-Device

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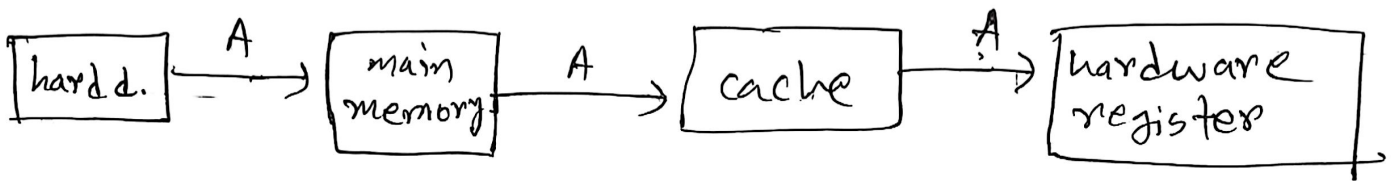


Caching

Concept: Data currently in use is temporarily copied from slower main storage to faster memory to speed up overall execution.

Migration of Data 'A' from Disk to Register

Multitasking environments must be careful to use most recent value, no matter where it is stored in the storage hierarchy.



Multiprocessor environment must provide cache coherency in hardware such that all CPUs have the most recent value in their cache.

Cache Coherency: Cache coherency helps to prevent cache inconsistency. MESI algorithm is an example used in cache coherency.

Computer-System Architecture

Most system use a single general-purpose processor.

Single vs Multiprocessor. Most legacy computers use one main CPU, but multiprocessor also known as parallel system or tightly coupled systems are widely used today.

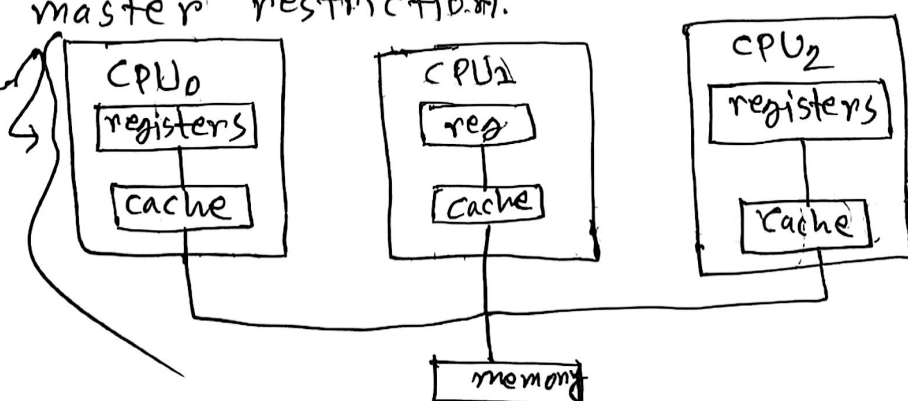
Key Advantages ① Increased Throughput: Accomplishes more work in less time.

② Economy of Scale: Sharing hardware resources costs less than separate standalone computers.

③ Increased Reliability: If one processor fails, the system continues running at a reduced pace (graceful degradation).

Two types Asymmetric Multiprocessing: Each processor is assigned a specific task.

Symmetric Multiprocessing. Every processor performs all tasks without a master restriction.



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Dual-Mode Operation

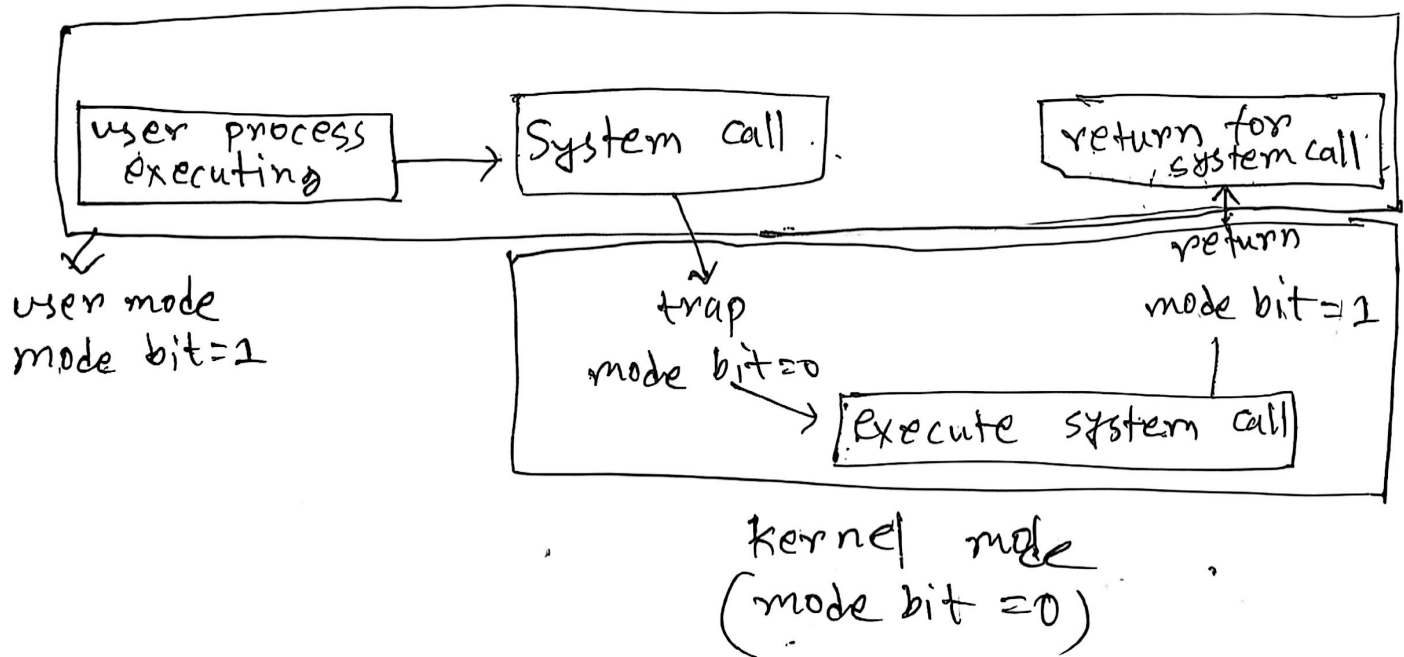
Purpose: Allows the operating system to protect itself and other running programs from bad user applications.

Hardware modes:

User mode (Mode bit=1): Executes standard user application code ~~but~~ with restricted privileges

Kernel Mode (Mode bit=0): Executes core OS code with full access to hardware; privileged instructions run only here.

Mode Switch: A user process makes a System Call;



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